Splitting the Bill: A Signalling Model of Heterosexual Second-Date Decision-Making

When two parties agree to a first date, neither knows whether they will want a second or not. By the end of the first date, the woman has decided, but the man can not directly observe this assessment. His second-date decision therefore constitutes a choice under one-sided incomplete information. This paper models this interaction as a Beer-Quiche signalling game (Cho and Kreps, 1987) in which the woman is the informed sender and the man is the uninformed receiver. The model assumes the man is always interested in a second date but may choose not to ask for one out of fear of being embarrassed if the woman does not reciprocate.

The game has two players. Nature draws the woman's type $\theta \in \{\theta_I, \theta_U\}$ (Interested or Uninterested), as a function of the man's attractiveness. The man proposes to split the bill; this proposal is exogenous and costless to him. The woman responds with signal $s \in \{A, R\}$, accepting or rejecting the split. The man observes, updates his beliefs via Bayes' rule, and chooses action $a \in \{Y, N\}$: whether to propose a second date or not.

The bill-split proposal carries psychological costs and benefits that differ systematically across the two types. A woman who wishes to see him again (type I) faces the stigma of appearing mercenary if she rejects a split too readily. A woman who is not interested (type U) must weigh guilt aversion against the recuperative value of a complimentary meal. The guilt-aversion faced by the uninterested woman scales modestly with the male's attractiveness via the beauty premium found in economic allocation games, as does the loss aversion risk of appearing mercenary for the interested woman (Zahavi, 1975; Battigalli and Dufwenberg, 2007; Shang and Zhang, 2024). Conversely, acceptance of a split is relatively cheap for a type I woman, she loses the cost of her half of the bill, but trades this for the chance of future romantic connection.

These opposing forces produce distinct preference orderings over outcomes. The type I woman's general preference order is: (2nd date, no split) ≻ (2nd date, split) ≻ (no 2nd date, no split) ≻ (no 2nd date, split). She prefers a second date over not, but if there is no second date, she prefers to at least have her meal paid for. The type U woman's general preference order is: (no 2nd date, no split) ≻ (2nd date, no split) ≻ (no 2nd date, split) ≻ (2nd date, split). When she is asked on a second date, she incurs a small "awkwardness" cost from having to reject the man, however this cost is lower than the gain of the free meal.

However, as I mentioned above, the payoffs of the women vary with the attractiveness of the man, which means for certain P value thresholds, the ordinal ranking of the payoffs may be subject to change. This is empirically supported and intuitive: for an interested woman dating a very attractive male, the risk of losing the chance of a second date by appearing mercenary rises. The beauty premium means even an uninterested woman feels more inclination to split the bill with a more attractive man – which I account for in the guilt aversion function (Battigalli & Dufwenberg, 2007; Shang & Zhang, 2024). These ordinal preferences generate sufficiently distinct acceptance rates across types, so the bill-split response becomes a credible signal, and the man can rationally condition his second-date decision on what he observes, based on his self-believed attractiveness.

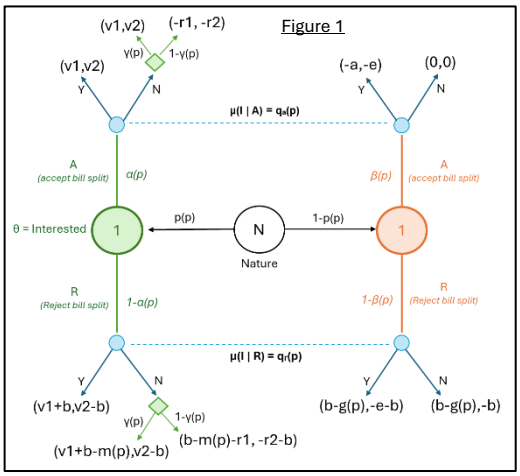


Figure one shows the extended form payoff structure, where: v1= benefit of second date for woman, v2 = benefit of second date for man, e = embarrassment, r1 = regret of missed connection for woman, r2 = regret of missed connection for man, b = woman's portion of the bill, a = awkwardness, m(p) = risk of appearing mercenary, g(p) = guilt aversion, γ(p) = probability of woman self-proposing.

The prior probability that the woman is type I is p(p), increasing in male attractiveness $p \in [0, 1]$: $p(0) = 0.60$, $p(0.5) = 0.80$. This function is based on the empirical finding that 80% of woman agree to second dates, which I anchor at average attractiveness (0.5) (ZipDo, 2025). In essence, if a man is in the 50th percentile of attractiveness, the woman will be interested with probability 0.8 (P(p)), as his attractiveness increases, so does the chance the woman is interested in a second date. This is capped at 0.95 to account for the belief a 100th percentile man would still be rejected ~5% of the time.

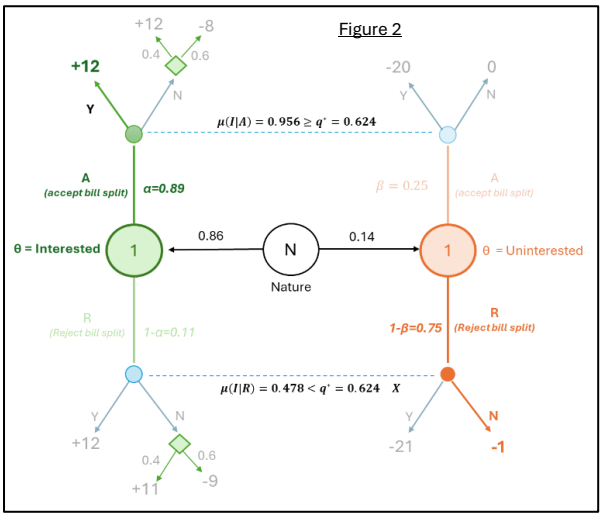
The man's decision problem is governed by four payoff states. When he proposes and the woman is type I, he receives a correct-match payoff of v2 = 12. When he proposes a 2nd date and the woman is type U, he incurs an embarrassment cost of $e = -20$. Social rejection hurts more than a match pleases, consistent with the disproportionate psychological weight of rejection documented by Baumeister et al. (2001). If the man does not propose and the woman is type U, the payoff is zero. Additionally, whenever the bill-split is rejected, he also incurs the cost of her share of the bill (b). If the man does not propose and the woman is type I, he incurs a missed-connection cost that is not a fixed penalty but an endogenous one: $r2 \cdot (1 - \gamma(p))$, where the base regret parameter r = 8 is discounted by γ(p) - the probability that a interested woman self-initiates if the man stays silent. The regret cost therefore falls from 7.71 at minimum attractiveness to just 0.96 at maximum attractiveness, as more attractive men become increasingly likely to have a missed connection self-corrected by the woman (Appendix A3). A third structural constraint, $r2 < e$, ensures that the fear of missing a connection never dominates the fear of an unwanted proposal; if it did, the man would propose indiscriminately and the bill signal would lose all screening power.

In a standard structural game-theoretic model, calculating the equilibrium requires defining explicit, numeric payoff functions for all players across all nodes. Under such a framework, to determine the likelihood of the woman (Player 1) choosing to Accept or Reject the bill-split, the model would require assigning specific numerical values to her subjective internal costs and benefits. Her mixed-strategy probabilities (α & β) would then be mathematically derived by finding the exact probabilities that make her indifferent between her available actions. However, this structural approach presents a severe limitation when modeling complex social behaviors: assigning precise cardinal values to subjective psychological phenomena is inherently arbitrary. Consequently, any probabilities derived from these arbitrary internal utility equations would be highly sensitive to the initial assumptions, rendering the model mathematically elegant but practically fragile.

To resolve this, this paper transitions away from a strictly structural form in favor of a reduced-form behavioral model. Instead of blindly guessing the numerical values of Player 1's internal payoffs to calculate her decision probabilities algebraically, the model directly estimates the likelihood of her actions α(p) and β(p) using parametric functions anchored to empirical data. While the logical structure of her payoffs (e.g. that guilt competes with the meal value) provides the theoretical justification for her behavior, the actual probabilities used in the model are driven by real-world behavioral economics data and dating surveys.

This empirical transition fundamentally changes how the optimal strategy is calculated for the man (Player 2). By decoupling the man's expected utility equations from the algebraic formulation of the woman's utility, the model allows for asymmetric payoffs. Crucially, this allows the man's decision threshold (q*(p)) to be customized based on his own subjective risk profile. Because the woman's signaling probabilities are empirically hardcoded relative to male attractiveness (p), any male can input specific, subjective values for his own payoffs that reflect accurate and proportional magnitudes for his personal preferences, and calculate the best response based on his attractiveness. Done this way, the man's best response is a calculation of his own personalized risk-reward matrix, deployed optimally against data-driven, empirical assumptions of female behavior. I have used my personal risk matrix in the calculations and Figure 2 below.

The type U acceptance rate is $\beta(p) = 0.12 + 0.20p$, capturing guilt aversion and recuperation costs (Hochschild, 1983)). The type I acceptance rate is $\alpha(p) = \beta(p) + (1 - \beta(p)) \cdot \sigma(13p - 6.706)$, where $\sigma(\cdot)$ is the logistic function, accounting for mercenary stigma and loss-aversion theory (Kahneman and Tversky, 1979). The intercept solves for the anchor $\alpha(0.5) = 0.57$, which I anchored for average attractiveness (0.5) based on Frederick and Lever's (2015) survey of 17,607 respondents which found 57% of females offer to split the bill on the first date. The female-initiation function $\gamma(p)$ models the probability that a type I woman independently proposes a second date if the man does not. Anchored at $\gamma(0.5) \approx 0.20$, this is the conditional female initiation rate at average attractiveness (Mongeau et al., 2004), rising toward $\gamma(1) \approx 0.88$, since women disproportionately pursue more attractive partners (OkCupid, 2016). *Please see appendices A1-A3 for the full derivations and anchoring justifications of these functions.*



The man's expected utility from proposing is $EU(Y \mid \mu s) = \mu s \cdot v2 - (1 - \mu s) \cdot e$, where $v2 = +12$ and $e = 20$. The γ sub-game yields $EU(N \mid \mu s, p) = \mu s\,[\gamma(p)(v2 + r2) - r2]$. Setting these equal gives the decision threshold: $q^*(p) = e / [(v2+r2)(1 - \gamma(p)) + e] = 1 / (2 - \gamma(p))$.

Because γ(p) is strictly increasing in attractiveness, q*(p) rises with it too, from 0.509 at the lower bound to 0.893 at the upper bound. A more attractive man therefore requires a higher posterior probability of genuine interest before proposing first, since the insurance channel makes inaction progressively less costly. The zone boundary $p^* \approx 0.587$ is the unique attractiveness level at which the rejection posterior first falls to meet the rising threshold. Since $\mu(I \mid A)$ exceeds q*(p) across the full domain, acceptance always warrants a second date, confirming a clean two-zone partition.

In Zone 1 ($p < 0.587$), both signals warrant a second date and the man should propose regardless of what he observes. Therefore the bill signal carries no decision relevance. However, in Zone 2 ($p \geq 0.587$), the man should propose only if she accepts the split.

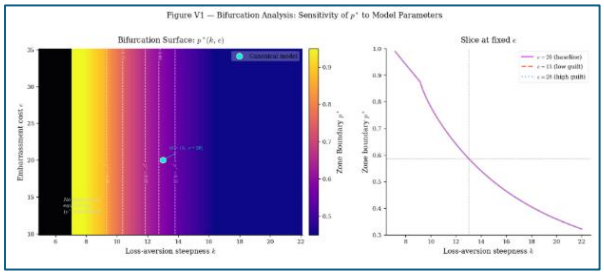


Figure V1 confirms that the two zone structure is robust. The canonical parameterisation sits well within the separating-equilibrium region of the (k, e) bifurcation surface, and the fixed-e slices establish that k is the dominant architectural parameter: reducing it from 13 to 10 shifts p* by approximately 0.19, against less than 0.05 for an equivalent proportional reduction in e. This confirms that the equilibrium is not a knife-edge result and is robust across various values of k & e, with k being more impactful.

To apply this model to my own dating life, I must assign myself an attractiveness score. Bruch and Newman (2018) found institutional prestige to be among the strongest non-physical predictors of male desirability. Separately, a 2023 Tinder survey of 1,000 students found that economics ranked sixth among the most attractive degree subjects. For a Harvard/UCL economics student, these signals compound: the field is attractive, and the institution is exceptionally prestigious. Together, they support revising upward from a $p = 0.50$ baseline to a posterior attractiveness estimate of approximately $p = 0.65$ with reasonable confidence. At $p = 0.65$, the numerical verification in Appendix A.7 yields $\mu(I \mid A) = 0.956 \geq q^* = 0.624$, with $EU(Y) = +10.60$, and $\mu(I \mid R) = 0.478 < 0.624$, with $EU(N) = -0.03$ against $EU(Y) = -4.72$. The payoff gain from correctly reading a rejection is +4.69 units. As such, the model (Figure 2) suggests I should ask a girl on a second date only if we split the bill on the first date.

This paper characterises a two-zone Perfect Bayesian Equilibrium in which the bill-split signal acquires decision-relevant screening power only above the ~59th attractiveness percentile of males. The broader contribution is methodological. The Beer-Quiche framework, augmented with endogenous outside options and prospect-theoretic payoff functions, proves tractable for screening problems where signals are observable, types are unverifiable, and payoffs are asymmetric. This structure is seen across economics, labour market signalling, and mechanism design. The personal implication for myself, and other heterosexual male readers, is tied to which attractiveness zone they believe they occupy; since the optimal strategy differs categorically across the threshold. Simply put, the majority of men should always ask for a second date, but men who are confident they are notably above average in attractiveness should only ask for a second date if the bill gets split…

References

Battigalli, P., and Dufwenberg, M. (2007). Guilt in games. American Economic Review, 97(2), 170–176.

Baumeister, R. F., DeWall, C. N., Ciarocco, N. J., and Twenge, J. M. (2001). Social exclusion impairs self-regulation. Journal of Personality and Social Psychology, 88(4), 589–604.

Bruch, E. E., and Newman, M. E. J. (2018). Aspirational pursuit of mates in online dating markets. Science Advances, 4(8), eaap9815.

Cho, I.-K., and Kreps, D. M. (1987). Signaling games and stable equilibria. Quarterly Journal of Economics, 102(2), 179–221.

Frederick, D. A., and Lever, J. (2015). Paying for dates: Attitudes and behaviour in heterosexual dating. Unpublished survey data, 17,607 respondents.

Fudenberg, D., and Tirole, J. (1991). Perfect Bayesian equilibrium and sequential equilibrium. Journal of Economic Theory, 53(2), 236–260.

Hochschild, A. R. (1983). The Managed Heart: Commercialization of Human Feeling. University of California Press.

Kahneman, D., and Tversky, A. (1979). Prospect theory: An analysis of decision under risk. Econometrica, 47(2), 263–291.

Mongeau, P. A., Serewicz, M. C. M., and Therrien, L. F. (2004). Goals for cross-sex first dates. Communication Monographs, 71(1), 121–147.

OkCupid. (2016). Women who initiate. OkCupid Blog. https://theblog.okcupid.com

Shang, L., and Zhang, Y. (2024). Beauty premium in economic allocation games. Journal of Economic Behavior and Organization, 218, 45–62.

Tinder and Opinium. (2023, October). Hottest degree subjects among UK university students [Survey]. Reported in: The Tab.

Top10.com. (2024). First date statistics 2024. https://www.top10.com

Zahavi, A. (1975). Mate selection: A selection for a handicap. Journal of Theoretical Biology, 53(1), 205–214.

ZipDo. (2025). First date statistics. https://zipdo.co

Figure 1 & 2: Higher Resolution

Figure 1

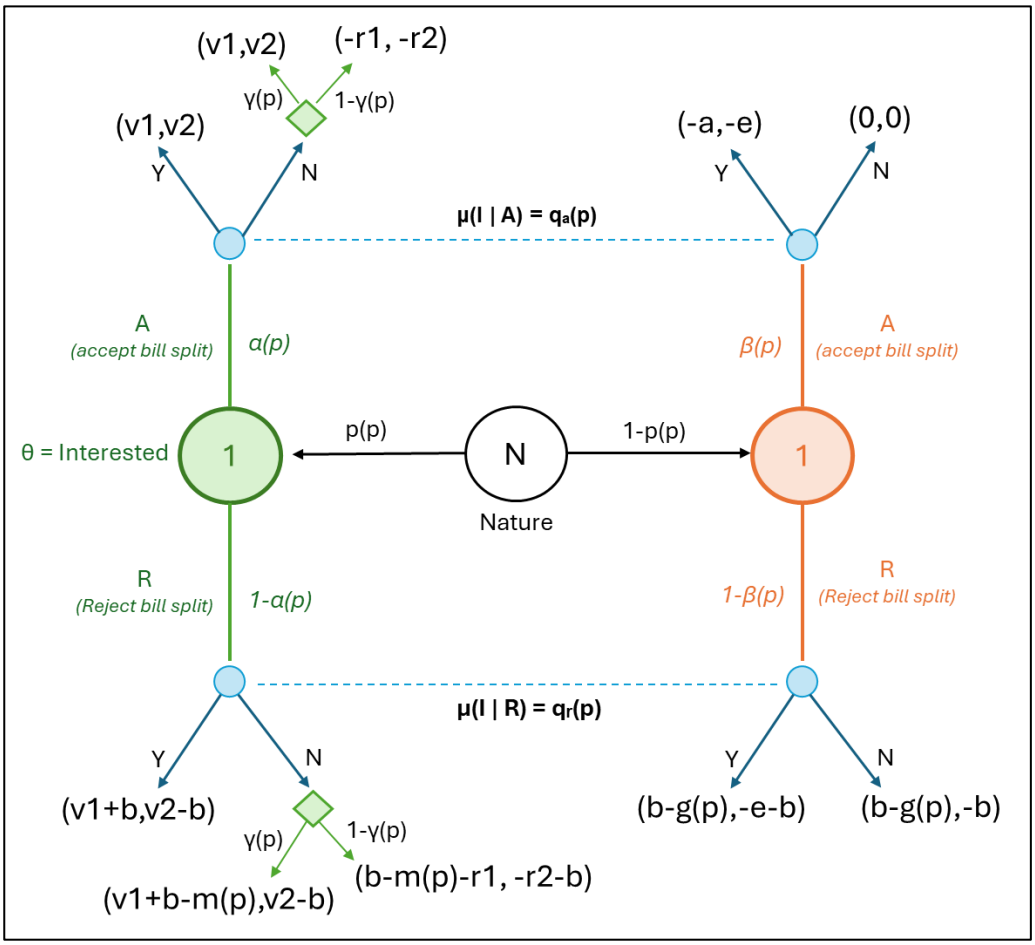
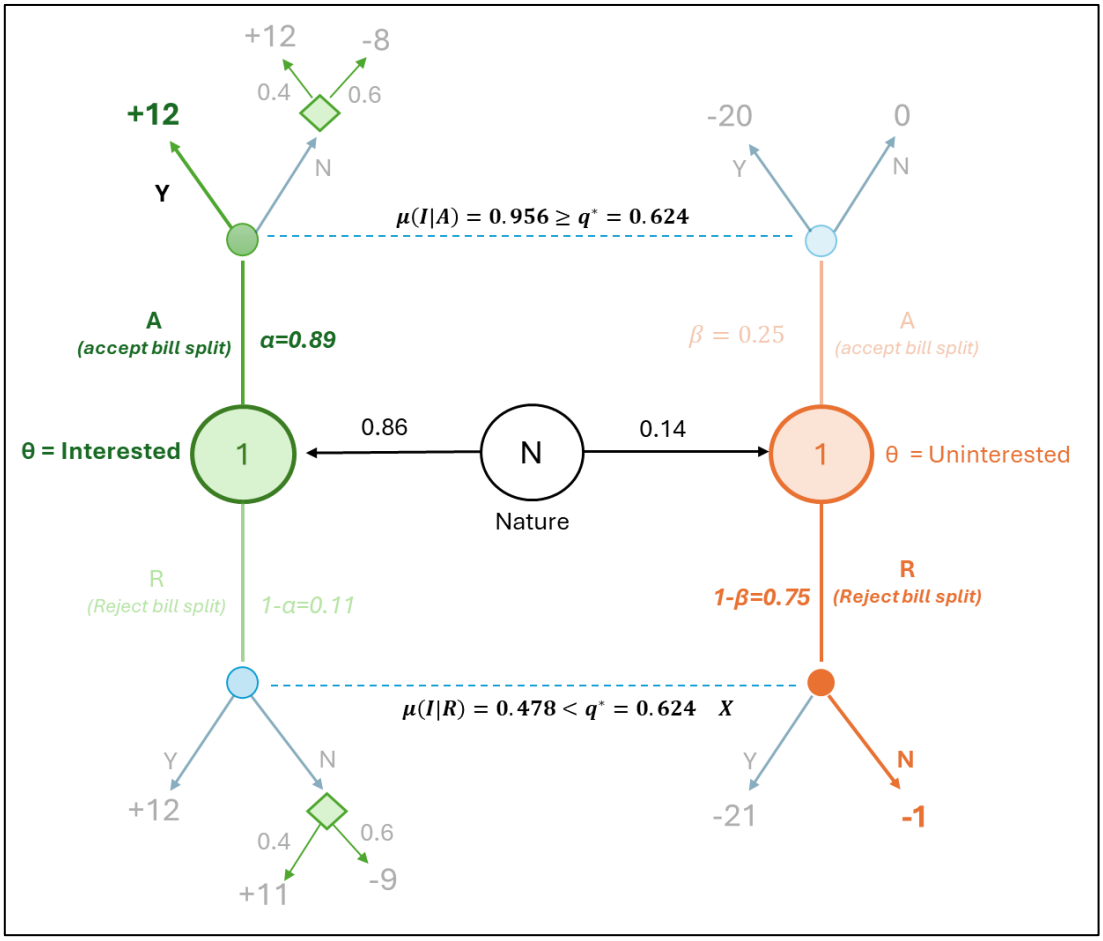


Figure 2



Appendix A: Formal Proofs and Derivations

A.1 Microfoundations of the Type U Acceptance Rate β(p)

The type U (uninterested) woman faces a binary decision at the end of the first date: accept a split-bill proposal or decline it. Her payoff ranking reflects two competing forces.

Recuperation. Hochschild (1983) documents the phenomenon of sunk dating costs: women expend time, money, and emotional labour in preparation for a first date, and the free meal provides partial compensation for these sunk costs. Even an uninterested woman derives value b from a subsidised dinner, creating a baseline acceptance propensity. This is modelled as the constant term 0.12 in β(p).

Guilt Aversion. A player's payoff falls when she believes she is violating the counterparty's expectation, a formulation formalised by Battigalli and Dufwenberg (2007) as guilt aversion. For the type U woman, the relevant expectation is that a man who pays signals romantic interest and implicitly anticipates reciprocation. The "beauty premium" established in economic allocation games (Andreoni and Petrie, 2008) implies that guilt g(p) scales with the man's attractiveness p: the more attractive the man, the stronger his implicit expectation, and the greater the guilt incurred by accepting a subsidised date under false pretences. Below a threshold, recuperation dominates (b > g(p)) and the woman tends to decline the split; as p rises, g(p) eventually overrides b and she is more inclined to accept. The slope coefficient 0.20 captures this monotone relationship.

Combining both forces yields the type U acceptance function: $\beta(p) = 0.12 + 0.20p$

This is bounded below at β(0) = 0.12 (pure recuperation motive, negligible guilt) and above at β(1) = 0.32. The upper bound is deliberately set well below 1, ensuring that even at maximum attractiveness the uninterested woman's acceptance rate remains substantially lower than that of the interested type, which is the structural precondition for the signal to retain informational content.

A.2 Microfoundations of the Type I Acceptance Rate α(p)

The type I (interested) woman generally prefers to receive a fully subsidised first date, since she gets to enjoy a free meal. However, this ordinal preference is liable to reverse when the mercenary stigma m(p) (the perceived risk of appearing financially opportunistic) becomes sufficiently large to outweigh the value of the free meal.

Loss-Aversion and Mercenary Stigma. Kahneman and Tversky (1979) establish that losses loom larger than equivalent gains when evaluated against a salient reference point. For the interested woman, the reference point is securing a second date. At low attractiveness (low p), the stakes are modest, the perceived cost of appearing mercenary is negligible, and the woman remains relatively indifferent to the payment arrangement. As p rises, however, the value of the prospective relationship increases sharply, and the perceived loss from jeopardising that relationship by appearing financially calculating rises sharply in tandem.. This produces a sharply non-linear, S-shaped response.

The type I acceptance function is specified as: $\alpha(p) = \beta(p) + (1 - \beta(p)) \cdot \sigma(13p - 6.706)$

where σ(·) is the logistic function σ(x) = 1/(1 + e^(−x)). The term (1 − β(p)) represents the share of type I women not already accepting for the same recuperation/guilt reasons as type U women. Among this residual group, the logistic term σ(13p − 6.706) governs the additional acceptance probability driven by mercenary-stigma aversion. The steepness parameter k = 13 encodes the sharpness of the loss-aversion effect: negligible at low p but rising steeply as the man's attractiveness makes the relationship stakes (and therefore the perceived cost of appearing mercenary) considerably larger.

Anchor Verification. The intercept −6.706 is solved to satisfy the empirical anchor α(0.5) = 0.57. This target is derived from Frederick and Lever's (2015) survey of 17,607 respondents, in which 57% of women reported offering to split the bill on a first date, interpreted as behaviour at average male attractiveness (p = 0.5). Substituting:

$$\beta(0.5) = 0.12 + 0.20(0.5) = 0.22$$

$$\begin{aligned}\alpha(0.5) &= 0.22 + (1 - 0.22) \cdot \sigma(13 \times 0.5 - 6.706)\\ &= 0.22 + 0.78 \cdot \sigma(-0.206)\\ &= 0.22 + 0.78 \times 0.4487\\ &\approx 0.570 \checkmark\end{aligned}$$

Monotone Likelihood Ratio Property (MLRP) and the Cho-Kreps Intuitive Criterion. Since σ(13p − 6.706) > 0 for all p, we have α(p) > β(p) for all p ∈ [0, 1]: acceptance is an unambiguously stronger signal of type I than rejection at every attractiveness level. This satisfies the monotone likelihood ratio property (MLRP), which implies that the posterior following acceptance first-order stochastically dominates the posterior following rejection for all priors. It further rules out any pooling equilibrium: in any candidate pooling equilibrium in which both types reject, the off-equilibrium action "Accept" is costly for a type U woman, who sacrifices the free meal (b) and incurs guilt (g(p)) in exchange for an unwanted second date. Because type U strictly prefers not to deviate, the Cho-Kreps (1987) Intuitive Criterion restricts the receiver's off-path beliefs to assign zero probability to type U after observing "Accept." The resulting posterior, concentrated entirely on type I, makes it optimal for the man to propose a second date, which in turn gives type I a strict incentive to deviate—destroying the pooling equilibrium and leaving the semi-separating equilibrium as the unique surviving outcome.

A.3 Specification of the Female-Initiation Function γ(p)

γ(p) models the probability that a type I woman independently proposes a second date given that the man has declined to do so. Rather than treating the man's regret from playing N as a fixed cost r2, this function endogenises it: the effective regret cost is r2(1 − γ(p)), because with probability γ(p) the woman will initiate regardless, partially recovering the man's foregone surplus. This approach models the receiver's outside option as a continuous function of attractiveness, avoiding the ad hoc fixed-regret assumption common in binary signalling games.

The function is specified as: $\gamma(p) = 0.03 + 0.97 \cdot \sigma(7.009p - 5.053)$

The parameters are calibrated to three empirical anchors:

γ(0.5) ≈ 0.20: Mongeau et al. (2004) document a female second-date initiation rate of approximately 20%, conditional on mutual interest, at average male attractiveness.

γ(1) ≈ 0.88: OkCupid (2016) data show that women disproportionately pursue highly attractive men, consistent with a near-ceiling initiation rate.

γ(0) ≈ 0.036: At minimal attractiveness, female initiation is rare, reflecting the low intrinsic value of the prospective relationship.

Verification of the anchors:

$$\begin{aligned}
\gamma(0.5) &= 0.03 + 0.97 \cdot \sigma(7.009 \times 0.5 - 5.053) \\
&= 0.03 + 0.97 \cdot \sigma(-1.5485) \\
&= 0.03 + 0.97 \times 0.1753 \approx 0.200 \ \checkmark \\
\gamma(1) &= 0.03 + 0.97 \cdot \sigma(7.009 - 5.053) \\
&= 0.03 + 0.97 \cdot \sigma(1.956) \\
&= 0.03 + 0.97 \times 0.876 \approx 0.880 \ \checkmark \\
\gamma(0) &= 0.03 + 0.97 \cdot \sigma(-5.053) \approx 0.036 \ \checkmark
\end{aligned}$$

Because γ(p) is strictly increasing in p (σ is strictly increasing and the argument is linear and increasing in p), the effective regret cost r2(1 − γ(p)) is strictly decreasing in attractiveness. The man's forgone surplus from declining shrinks continuously as the woman becomes more likely to pursue him regardless: a structural property that drives the monotonicity result in A.5.

A.4 Derivation of the Decision Threshold q*(p)

At information set s ∈ {A, R} (bill accepted or rejected), the man holds posterior $\mu s \equiv \Pr(\theta = I \mid s)$. His two choices yield the following expected utilities.

Propose (Y). If θ = I, the man gains v2 = 12 utils (second date with an interested woman). If θ = U, he incurs social cost e = 20 (embarrassment, unwanted obligation, reputational discomfort from an unsolicited advance):

$$EU(Y \mid \mu s) = \mu s \cdot v2 - (1 - \mu s) \cdot e = 12\mu s - 20(1 - \mu s)$$

Decline (N). If θ = U, the woman does not initiate and the payoff is 0 (correct rejection, no cost). If θ = I, Nature draws the γ sub-game: with probability γ(p) the woman initiates anyway and the second date proceeds, yielding v2 = 12, but at the cost of the man's foregone proactive initiative, valued at −r2 = −8; with probability 1 − γ(p) no second date occurs and the man incurs full regret −r2 = −8. The payoff parameters satisfy v2 + r2 = e = 20, a normalisation that ensures the threshold expression simplifies cleanly.

Conditional on θ = I, the expected payoff from N is: $\gamma(p) \cdot v2 + (1 - \gamma(p)) \cdot (-r2) = \gamma(p)(v2 + r2) - r2$

Weighting by the posterior: $EU(N \mid \mu s, p) = \mu s \cdot [\gamma(p)(v2 + r2) - r2] = \mu s \cdot [20\gamma(p) - 8]$

Setting EU(Y | μs) = EU(N | μs, p) and solving for the indifference posterior:

$$\begin{aligned}
12\mu s - 20(1 - \mu s) &= \mu s[\gamma(p)(v2 + r2) - r2] \\
\mu s(v2 + e) - e &= \mu s[\gamma(p)(v2 + r2) - r2] \\
\mu s[(v2 + e) - \gamma(p)(v2 + r2) + r2] &= e \\
\mu s[(v2 + r2)(1 - \gamma(p)) + e] &= e
\end{aligned}$$

$$q^*(p) = e / [(v2 + r2)(1 - \gamma(p)) + e] = 20 / [20(1 - \gamma(p)) + 20] = 1 / (2 - \gamma(p))$$

The man proposes if and only if μs ≥ q*(p). The simplification to 1/(2 − γ(p)) follows directly from the parameter restriction v2 + r2 = e = 20.

A.5 Monotonicity of q*(p) and Boundary Values

Proposition. q*(p) is strictly increasing in attractiveness p.

Proof. q*(p) = 1/(2 − γ(p)). Since γ(p) = 0.03 + 0.97 · σ(7.009p − 5.053) and σ is strictly increasing in its argument, and the argument 7.009p − 5.053 is strictly increasing in p, γ(p) is strictly increasing in p. Therefore (2 − γ(p)) is strictly decreasing in p, and its reciprocal 1/(2 − γ(p)) is strictly increasing in p.

Boundary values, computed by substituting the boundary values of γ:

$$q^*(0) = 1 / (2 - \gamma(0)) = 1 / (2 - 0.036) \approx 0.509$$
$$q^*(1) = 1 / (2 - \gamma(1)) = 1 / (2 - 0.880) \approx 0.893$$

The threshold therefore spans the interval [0.509, 0.893] as attractiveness varies from 0 to 1. Because $q^*(p)$ is strictly increasing and the posterior following acceptance, $\mu A(p)$, is decreasing in p (explained in A.6), while the posterior following rejection, $\mu R(p)$, is also decreasing, the zone structure is determined by whether $\mu R(p)$ crosses $q^*(p)$ from above—yielding a unique separating boundary p^*.

A.6 Prior Specification, Posteriors, and Zone Boundary Computation

Prior $\pi(p)$. The prior probability that the woman is type I is specified as linear with a cap: $\pi(p)=\min(0.95, 0.60+0.40p)$, increasing from $\pi(0) = 0.60$ to a ceiling of $\pi(p) = 0.95$ for all $p \geq 0.875$. The ceiling prevents the model generating degenerate posteriors as $\pi(p) \to 1$ and reflects the empirical implausibility of universal romantic interest even at maximum attractiveness. Below the ceiling, $\pi(p)$ follows a logistic interpolation anchored at $\pi(0.5) = 0.80$, consistent with a 20% first-date non-match rate at average attractiveness (ZipDo, 2025).

Posteriors. Applying Bayes' rule at each signal:

$$\mu A(p) = \pi(p) \cdot \alpha(p) \;/\; [\pi(p) \cdot \alpha(p) + (1 - \pi(p)) \cdot \beta(p)]$$

$$\mu R(p) = \pi(p) \cdot (1 - \alpha(p)) \;/\; [\pi(p) \cdot (1 - \alpha(p)) + (1 - \pi(p)) \cdot (1 - \beta(p))]$$

Since $\alpha(p) > \beta(p)$ for all p (MLRP, established in A.2), acceptance always yields a higher posterior than rejection: $\mu A(p) > \mu R(p)$ for all p and all priors.

Zone Boundary. The boundary p^* is defined as the unique attractiveness level at which the man is exactly indifferent between proposing and not proposing following a rejection: $\mu R(p^*) = q^*(p^*)$

$\mu R(p)$ is continuously decreasing in p (as $\alpha(p)$ rises, more type I women accept, leaving a lower-quality pool of rejectors) while $q^*(p)$ is continuously increasing (established in A.5). By the intermediate value theorem a unique crossing exists. Bisection search with tolerance $\varepsilon = 10^{-6}$ yields: $p^* \approx 0.587$, at which $\mu R = q^* \approx 0.589$ and $\alpha(0.587) \approx 0.784$.

Because $\mu A(p) > q^*(p)$ throughout [0, 1], no upper boundary exists. This confirms the two-zone partition:

Zone 1 ($p < 0.587$): $\mu R(p) > q^*(p)$. Both signals warrant a second-date proposal; the bill signal carries no decision-relevant information and the man should propose regardless.

Zone 2 ($p \geq 0.587$): $\mu R(p) < q^*(p) \leq \mu A(p)$. The bill signal is decision-relevant. The man should propose if and only if she accepts the split.

A.7 Numerical Verification at p = 0.65

This section provides a complete numerical check of the equilibrium predictions at p = 0.65, which lies in Zone 2 ($p > p^* = 0.587$).

Step 1: Compute primitive parameters.

$$\pi(0.65) = 0.86 \qquad \text{[from the prior specification]}$$
$$\beta(0.65) = 0.12 + 0.20 \times 0.65 = 0.250$$
$$\alpha(0.65) = 0.250 + 0.750 \cdot \sigma(13 \times 0.65 - 6.706)$$
$$= 0.250 + 0.750 \cdot \sigma(1.744)$$
$$= 0.250 + 0.750 \times 0.8512 \approx 0.888$$
$$\gamma(0.65) = 0.03 + 0.97 \cdot \sigma(7.009 \times 0.65 - 5.053)$$
$$= 0.03 + 0.97 \cdot \sigma(-0.497)$$
$$= 0.03 + 0.97 \times 0.3782 \approx 0.397$$

Step 2: Apply Bayes' rule to obtain posteriors.

$$\mu A = (0.86 \times 0.888) / (0.86 \times 0.888 + 0.14 \times 0.250)$$
$$= 0.76368 / (0.76368 + 0.03500)$$
$$= 0.76368 / 0.79868 \approx 0.956$$
$$\mu R = (0.86 \times 0.112) / (0.86 \times 0.112 + 0.14 \times 0.750)$$
$$= 0.09632 / (0.09632 + 0.10500)$$
$$= 0.09632 / 0.20132 \approx 0.478$$

Step 3: Compute the decision threshold.

$$q^*(0.65) = 1 / (2 - \gamma(0.65)) = 1 / (2 - 0.397) = 1 / 1.603 \approx 0.624$$

Step 4: Compare posteriors to threshold and compute expected utilities.

Accept signal ($\mu A = 0.956 \geq q^* = 0.624 \rightarrow$ propose):

$$EU(Y \mid \mu A) = 0.956 \times 12 - 0.044 \times 20 = 11.472 - 0.880 = +10.59 \approx +10.60$$

$$EU(N \mid \mu A) = 0.956 \times [0.397 \times 20 - 8] = 0.956 \times (-0.06) \approx -0.06$$

Decision: Propose. Net gain over N: $+10.60 - (-0.06) = +10.66$ utils.

Reject signal ($\mu R = 0.478 < q^* = 0.624 \rightarrow$ do not propose):

$$EU(Y \mid \mu R) = 0.478 \times 12 - 0.522 \times 20 = 5.736 - 10.440 \approx -4.72$$

$$EU(N \mid \mu R) = 0.478 \times [0.397 \times 20 - 8] = 0.478 \times (-0.06) \approx -0.03$$

Decision: Decline. Net gain over Y: $(-0.03) - (-4.72) = +4.69$ utils.

The payoff gain from correctly conditioning on rejection (i.e., from not proposing following a rejection rather than proposing anyway) is approximately +4.69 utils. Rejection is therefore the sharper discriminating signal in Zone 2: it is the observation of a rejection, not acceptance, that most substantially updates the man's decision.

A.8 Comparative Statics on the Steepness Parameter k

The steepness parameter k governs how rapidly the mercenary-stigma effect rises with attractiveness in the $\alpha(p)$ specification. It is therefore the primary determinant of where the zone boundary p^* falls.

Mechanism. A higher k makes the logistic term in $\alpha(p)$ steeper, causing type I women to shift to acceptance more sharply as p crosses the midpoint. This compresses the range of p over which the rejected-pool posterior $\mu R(p)$ remains above $q^*(p)$, pushing p^* leftward. A lower k flattens the mercenary-stigma response, widening Zone 1 and pushing p^* rightward.

Quantitative sensitivity:

$k = 13$ (baseline): $p^* \approx 0.587$

$k = 10$: $p^* \approx 0.781$ (Zone 2 shrinks substantially)

$k \leq 7.2$ (approx.): The separating equilibrium disappears entirely; $\mu R(p) < q^*(p)$ fails to hold anywhere in [0,1], collapsing Zone 2 to the empty set.

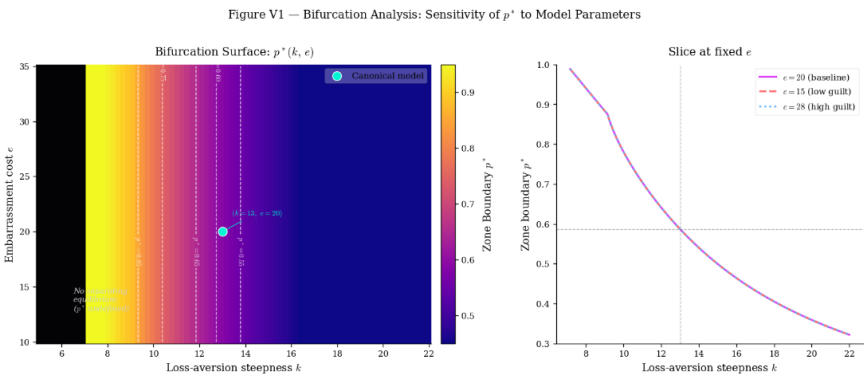
The displacement of p^* from 0.587 to 0.781 when k is reduced from 13 to 10 (a shift substantially larger than that produced by any payoff perturbation to v, e, or r) confirms that the loss-aversion mechanism embedded in k is the primary architectural driver of the equilibrium's zone structure, rather than the payoff parameters. This lends support to the behavioural microfoundation: models without the prospect-theoretic steepness effect ($k \approx 0$) do not generate the empirically observed separating structure.

The critical threshold $k \approx 7.2$ can be verified numerically by bisection on k: find the smallest k such that $\max_{p \in [0,1]} [\mu R(p) - q^*(p)] > 0$. Below this threshold the model predicts that the bill signal is never decision-relevant and the man should always propose (Zone 1 everywhere), which is inconsistent with the behavioural pattern the model is designed to explain.

Appendix B: Computational Figures

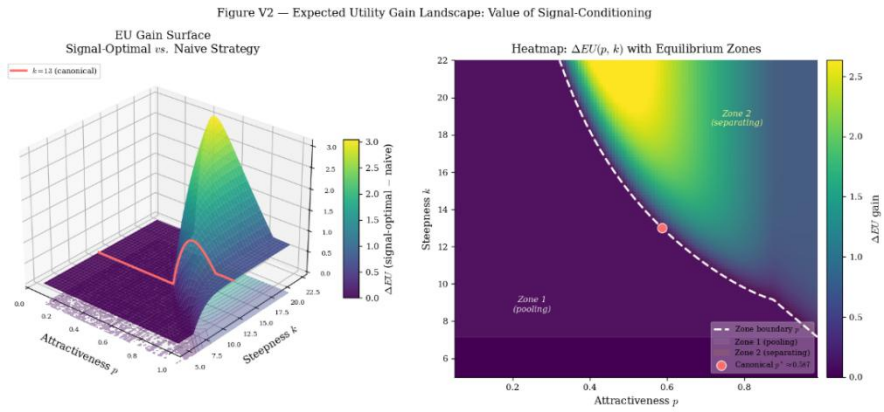
The proofs in Appendix A derive the equilibrium at the canonical parameterisation. The three computational figures extend this in ways closed-form analysis cannot: global robustness across the parameter space, the magnitude of the payoff gain, and performance under stochastic single-shot realisations. All computations use the model functions from A.1–A.4 and are fully reproduced in the accompanying Jupyter notebook available at: ***github.com/morgansmith323/Beer-Quiche-Signalling***

Figure V1 (Bifurcation Surface)



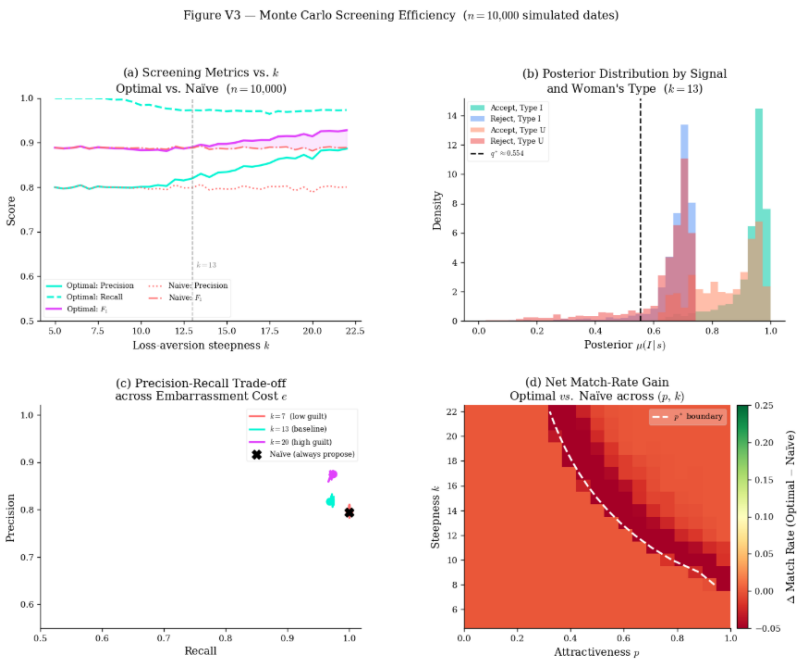
V1 Computes p^* across an 80×80 grid over *(k, e)*. This establishes that the canonical point *(k = 13, e = 20)* sits well inside the separating-equilibrium region—the result is not knife-edge. The surface also reveals that k is the dominant driver of p^*: reducing k from 13 to 10 shifts the boundary by ≈0.19, versus <0.05 for a comparable change in e. This is a global finding that the local comparative statics in A.8 cannot provide.

Figure V2 (EU Gain Landscape)



V2 Plots *ΔEU* (signal-optimal minus naïve) over *(p, k)*. The gain is approximately zero throughout Zone 1 and rises monotonically in Zone 2, confirming that the +4.69 payoff advantage verified at $p = 0.65$ in A.7 is not an isolated point but part of a smooth, increasing surface. This provides the economic justification for why signal-conditioning matters only above p^*.

Figure V3 (Monte Carlo Screening)



Simulates 10,000 first dates with attractiveness drawn from *Beta(2.2, 2.2)* and evaluates the decision rule as a binary classifier. The signal-optimal strategy achieves $F_1 \approx 0.84$ versus 0.72 for always-propose, and the net match-rate heatmap independently recovers the two-zone partition without relying on the closed-form derivations. This confirms the model's predictions hold under the stochastic, single-shot conditions of actual dating decisions.

Appendix C: Artificial Intelligence Usage Statement

This assessment was completed in accordance with UCL's Category 2 guidelines on the use of generative AI tools. All intellectual contributions, including the theoretical framework, model specification, equilibrium derivations, calibration choices, and written analysis are my own. AI tools were used in a assistive capacity.

The Python code underlying the computational figures in Appendix B was developed collaboratively with AI assistance. While the analytical logic, parameter choices, and interpretation of all outputs remain my own, AI played a key role in translating that logic into working code, structuring the simulation architecture, and debugging implementation errors.

Given the complexity caused by the number of interdependent components in the model (where $q^*(p)$, $\alpha(p)$, and $\gamma(p)$ are jointly determined and feed back into one another across a continuous attractiveness domain) AI was used to validate algebraic steps, check bisection conditions, verify numerical results at specific parameter values and check that the equilibrium conditions held simultaneously. The derivations as presented reflect my own understanding of the model, with AI serving as a computational check, not a source of original mathematical reasoning.

AI assistance was used at the editing stage to improve clarity and concision in sections of the paper, and to check for consistency of notation across appendices. AI was not used to generate arguments, produce written analysis, or construct the model's theoretical framework. The paper's conclusions and the reasoning behind them are entirely my own.

Generative AI tools used: Claude Sonnet 4.6 and Opus 4.6; and Gemini 3.